

Name: _____

Name the criterion (SSS, SAS, ASA, AAS, or HL) that the given information provides.

1. Three pairs of congruent sides.
2. Two pairs of congruent angles and the pair of congruent sides between them.
3. Two pairs of congruent sides and the pair of congruent angles between them.
4. Two right triangles with congruent hypotenuses and one pair of congruent legs.
5. Two pairs of congruent angles and a pair of congruent sides that is NOT between them.

Solve.

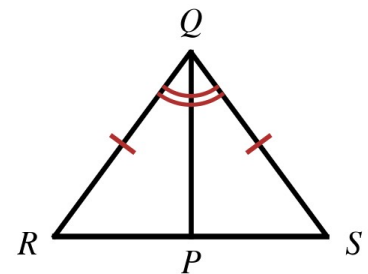
6. $\triangle ABC \cong \triangle DEF$, $AB = 3x + 2$, and $DE = 20$. Find x .
7. $\triangle MNO \cong \triangle PQR$, $MN = 7y - 5$, and $PQ = 30$. Find y and MN .

Answer each question.

8. Two triangles have two pairs of congruent sides and a pair of congruent angles that is NOT between those sides. Is that enough to prove them congruent? Explain.
9. How is HL different from SSA, given that HL also uses two sides and a non-included angle?

Complete the two-column proof.

- 10. Given:** $\overline{QP}$ bisects $\angle RQS$; $\overline{QR} \cong \overline{QS}$ **Prove:** $\triangle RQP \cong \triangle SQP$



Statements	Reasons
1. $\overline{QP}$ bisects $\angle RQS$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.